

Practice Exercise

User Defined Exceptions

Design and Implement a Java program as follows:

- 1) Use the Pet and IllegalPetNameArgumentException classes from class materials
- 2) Add a new class IllegalPetAgeArgumentException and modify Pet class to use it
- 3) Implement CreatePet class with main method which will prompts user for name and age and if either is incorrect, it will loop to prompt user again for both values. Catch the two user defined exception to determine if code should loop again.

SAMPLE RUN 1

```
Enter pet name:  
Enter pet age: 0
```

```
TRY AGAIN! Invalid pet name: null
```

```
Enter pet name: buz  
Enter pet age: 0
```

```
TRY AGAIN! Invalid pet age: Age must be greater than 0: age=0
```

```
Enter pet name: buz  
Enter pet age: 8
```

```
Successfully create Pet with name=buz age=8
```

SAMPLE RUN 2

```
Enter pet name: buz  
Enter pet age: 0
```

```
TRY AGAIN! Invalid pet age: Age must be greater than 0: age=0
```

```
Enter pet name:  
Enter pet age: 8
```

```
TRY AGAIN! Invalid pet name: null
```

```
Enter pet name: buz  
Enter pet age: 8
```

```
Successfully create Pet with name=buz age=8
```

Sample solution

```
public class Pet {

    private String name;
    private int age;

    public Pet(String name, int age) {

        if (name == null || name.isBlank() || name.isEmpty())
            throw new IllegalArgumentException(name);

        if (age <= 0)
            throw new IllegalArgumentException(age);

        this.name = name;
        this.age = age;
    }

    public String getName() {
        return name;
    }

    public int getAge() {
        return age;
    }

    // other methods here
}

public class IllegalPetNameArgumentException extends IllegalArgumentException {

    private String msg;

    public IllegalPetNameArgumentException(String name) {

        if (name == null)
            msg = "Pet name cannot be null";
        else if (name.isBlank())
            msg = "Pet name cannot have all blank values";
        else if (name.isEmpty())
            msg = "Pet name cannot be empty value";
    }

    @Override
    public String toString() {
        return this.getClass().getSimpleName() + " : " + msg;
    }
}

public class IllegalPetAgeArgumentException extends IllegalArgumentException {

    private int age;
```

```

    public IllegalPetAgeArgumentException(int age) {
        super("Age must be greater than 0: age=" + age);
    }

    @Override
    public String toString() {
        return this.getClass().getSimpleName() + " : Age must be greater than 0:
age=" + age;
    }
}

import java.util.Scanner;

public class CreatePet {

    public static void main(String[] args) {

        Scanner scanstr = new Scanner(System.in);
        Scanner scanint = new Scanner(System.in);

        boolean invalidValue = true; // Controls while loop for input
        String name = null;
        int age = 0;
        Pet pet = null;

        // loop until get valid name and age
        while (invalidValue) {

            // prompt for name
            System.out.print("\nEnter pet name: ");
            name = scanstr.nextLine();

            // prompt for age
            System.out.print("Enter pet age: ");
            age = scanint.nextInt();

            try {

                pet = new Pet(name, age); // create instance of Pet
                invalidValue = false; // constructor did not throw
exception so can exit loop

            } catch (IllegalPetNameArgumentException e) {
                System.out.println("\nTRY AGAIN! Invalid pet name: " +
e.getMessage());
            } catch (IllegalPetAgeArgumentException e2) {
                System.out.println("\nTRY AGAIN! Invalid pet age: " +
e2.getMessage());
            }
        }

        // code doing something with Pet instance

```

```
        System.out.println("\nSuccessfully create Pet with name=" +  
pet.getName() + " age="+pet.getAge());  
    }  
}
```